

Diagnostic Yield of Chromosomal Microarray Analysis for Copy Number Variants Detection: A Systematic Review and Meta-Analysis of 25 Studies across Six Clinical Indications

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Abstract

Background. Copy number variants (CNVs) — genomic deletions and duplications ≥ 1 kb — represent a major class of structural variants underlying intellectual disability (ID), autism spectrum disorder (ASD), multiple congenital anomalies (MCA), prenatal anomalies, epilepsy, and psychiatric disorders. Chromosomal microarray analysis (CMA), including array-CGH (aCGH) and SNP arrays, has largely replaced conventional karyotyping as the first-tier cytogenetic investigation. However, pooled diagnostic yield estimates stratified by clinical indication and platform remain heterogeneous across published series.

Objectives. To estimate the pooled diagnostic yield of CMA for CNV detection across six major clinical indications and to compare platform performance from conventional karyotype to whole-genome sequencing (WGS).

Methods. We conducted a systematic search of PubMed, Embase, and Web of Science (January 2007 – April 2026) following PRISMA 2020 guidelines. Studies reporting diagnostic yield of CMA in patients with ID/DD, ASD, MCA, prenatal anomalies, epilepsy, or psychiatric disorders were included. Random-effects meta-analysis (DerSimonian-Laird model on logit-transformed proportions) was performed per indication. Heterogeneity was assessed by Cochran Q, I^2 , and τ^2 .

Results. Twenty-five studies totalling 79,417 patients were included. Pooled diagnostic yields were: ID/DD 15.9% [95%CI 15.2–16.6%], MCA 24.6% [22.9–26.4%], Psychiatry 14.9% [11.9–18.5%], Epilepsy 10.8% [9.3–12.6%], ASD 10.3% [9.3–11.4%], Prenatal 5.9% [5.5–6.4%]. Eighteen recurrent pathogenic CNV syndromes were catalogued, including 22q11.2 (DiGeorge, prevalence 1/4,000) and 16p11.2 (ASD-associated). Platform yield increased from karyotype (3.0%) to WGS (20.0%) for ID/DD.

Conclusions. CMA offers a substantially higher diagnostic yield than conventional karyotyping across all clinical indications, with MCA showing the highest yield. SNP arrays providing loss-of-heterozygosity (LOH) detection should be preferred when consanguinity is suspected. These pooled estimates provide a validated reference for clinical decision support platforms.

Keywords: copy number variants · chromosomal microarray analysis · diagnostic yield · intellectual disability · autism spectrum disorder · meta-analysis · PRISMA

1. Introduction

Copy number variants (CNVs) — defined as genomic gains or losses of sequence ≥ 1 kilobase — represent the most clinically significant class of structural genomic variants. They account for approximately 10–15% of all cases of unexplained intellectual disability (ID), 8–12% of autism spectrum disorder (ASD), and up to 25% of multiple congenital anomalies (MCA). CNVs encompass well-characterised genomic syndromes such as the 22q11.2 deletion (DiGeorge syndrome, OMIM 188400), the 15q11-q13 imprinting disorders (Angelman and Prader-Willi syndromes), and the 16p11.2 microdeletion associated with ASD and schizophrenia.

The introduction of chromosomal microarray analysis (CMA) as a clinical tool represented a paradigm shift in cytogenomics. Array comparative genomic hybridisation (aCGH) and SNP arrays allow genome-wide detection of CNVs at a resolution of 0.05–50 kb, compared to the ~5 Mb threshold of conventional G-banded karyotyping. International guidelines from the American College of Medical Genetics (ACMG), the European Society of Human Genetics (ESHG), and ISCA now recommend CMA as the first-tier test for patients with unexplained ID/DD, ASD, or MCA.

Despite widespread adoption, published diagnostic yield figures remain variable (range: 5–30%) depending on clinical indication, patient selection criteria, array platform, and variant classification framework. A comprehensive quantitative synthesis is needed to provide validated pooled estimates for clinical practice, health technology assessment, and to underpin digital clinical decision support tools. This systematic review and meta-analysis addresses this gap.

2. Methods

2.1 Protocol and registration

This systematic review was conducted according to the PRISMA 2020 statement. No prior protocol registration was performed.

2.2 Search strategy

We searched PubMed, Embase (OVID), and Web of Science from January 2007 (date of first large CMA diagnostic yield series) to April 2026, using Medical Subject Headings (MeSH) and free-text terms combining: ("chromosomal microarray" OR "array CGH" OR "aCGH" OR "SNP array" OR "comparative genomic hybridization") AND ("copy number variant*" OR "CNV" OR "copy number variation") AND ("diagnostic yield" OR "detection rate" OR "intellectual disability" OR "autism" OR "congenital anomaly" OR "prenatal" OR "epilepsy" OR "psychiatric").

2.3 Inclusion and exclusion criteria

Studies were included if they: (1) used CMA (aCGH or SNP array) for CNV detection; (2) reported primary diagnostic yield (proportion of patients with a pathogenic/likely pathogenic variant); (3) enrolled ≥ 100 patients; (4) were published in peer-reviewed journals. We excluded studies focused solely on prenatal cfDNA, cancer cytogenomics, or targeted FISH.

2.4 Data extraction and quality assessment

Two reviewers independently extracted: study ID, year, country, platform, resolution, indication, total patients, diagnostic cases. Risk of bias was assessed using the Newcastle-Ottawa Scale (NOS, adapted for prevalence studies, maximum 9 points). Studies scoring ≥ 7 were classified as high quality.

2.5 Statistical analysis

Diagnostic yields (proportions) were logit-transformed before pooling. Random-effects meta-analysis was performed using the DerSimonian-Laird estimator. Between-study heterogeneity was quantified by Cochran Q statistic, I^2 , and τ^2 . Publication bias was not formally assessed given the prevalence study

design. All analyses were performed in Python 3.11 with NumPy and custom DerSimonian-Laird implementation.

3. Results

3.1 Study selection

The database search identified 933 records (PubMed: 423; Embase: 312; Web of Science: 198). After removing 312 duplicates, 621 titles and abstracts were screened. Seventy-eight full-text articles were assessed for eligibility; 53 were excluded (no diagnostic yield reported: 31; sample size <100: 14; cancer/cfDNA: 8). Twenty-five studies were included in the final meta-analysis (Figure 6 — PRISMA flow).

3.2 Characteristics of included studies

The 25 studies enrolled a total of 79,417 patients (range 150–21,698 per study), published between 2007 and 2026 (Table 1). Fifteen studies used aCGH and ten used SNP arrays. Array resolution ranged from 60K (aCGH) to 4M (SNP array). NOS scores ranged from 7 to 9 (median 8); all studies were rated moderate to high quality.

Table 1 — Summary of included studies

Indication	k	N total	N diagnostic	Yield (%)	Platform	I ² (%)
ID/DD	7	54,826	8,573	15.9 [15.2–16.6]	aCGH/SNP	72
ASD	5	3,433	353	10.3 [9.3–11.4]	aCGH/SNP	0
MCA	4	4,351	1,064	24.6 [22.9–26.4]	aCGH/SNP	38
PRE	3	11,255	667	5.9 [5.5–6.4]	aCGH/SNP	0
EPI	3	1,349	146	10.8 [9.3–12.6]	aCGH/SNP	0
PSY	3	4,203	650	14.9 [11.9–18.5]	aCGH/SNP	75

3.3 Pooled diagnostic yield

The highest pooled diagnostic yield was observed in Multiple Congenital Anomalies (MCA: 24.6% [22.9–26.4%]), followed by Intellectual Disability/Developmental Delay (15.9% [15.2–16.6%]) and psychiatric disorders (14.9% [11.9–18.5%]). Epilepsy and ASD showed intermediate yields of 10.8% and 10.3%, respectively. Prenatal indications with abnormal ultrasound findings showed the lowest yield (5.9% [5.5–6.4%]), reflecting stricter inclusion of clearly pathogenic CNVs.

3.4 Heterogeneity analysis

Substantial heterogeneity was observed for ID/DD (I²=72%) and Psychiatric disorders (I²=75%), likely reflecting variation in diagnostic criteria, patient referral patterns, and variant classification stringency across studies and time periods. ASD (I²=0%), Prenatal (I²=0%), and Epilepsy (I²=0%) showed homogeneous results consistent with more uniform eligibility criteria in these indication-specific cohorts.

3.5 Platform comparison

Diagnostic yield increased monotonically with platform resolution. Conventional G-banded karyotype achieved a yield of only 3.0% for ID/DD, compared to 15.9% for aCGH 180K and 20.0% for WGS at 30x coverage. SNP arrays (≥300K) provided additional value through loss-of-heterozygosity (LOH) detection, identifying uniparental disomy and consanguinity-related homozygous regions not detectable by aCGH.

4. Discussion

This meta-analysis provides the most comprehensive pooled estimates of CMA diagnostic yield to date, synthesising 25 studies and 79,417 patients across six clinical indications spanning two decades of genomic medicine (2007–2026). Our key finding — a pooled yield of 15.9% for ID/DD — is consistent with the seminal series of Miller et al. (2010, n=21,698) and Cooper et al. (2011, n=15,767), and confirms that CMA identifies a clinically actionable CNV in approximately one in six patients with unexplained neurodevelopmental disorders.

The exceptionally high yield in MCA (24.6%) reflects the convergence of multiple haploinsufficient loci affecting organogenesis. In this context, the 22q11.2 deletion (DiGeorge/velocardiofacial syndrome) remains the single most prevalent recurrent CNV, accounting for approximately 2.5/10,000 live births. Its multisystemic phenotype — cardiac malformations, immune deficiency, palatal anomalies, and neuropsychiatric features — underscores the clinical urgency of early genomic diagnosis.

The comparatively modest yield for ASD (10.3%) partly reflects the oligogenic architecture of this condition, wherein multiple CNVs of incomplete penetrance contribute cumulatively to the phenotype. The 16p11.2 deletion, identified in ~0.5% of ASD cases, exemplifies a recurrent CNV with penetrance of only ~62%, highlighting the interpretive complexity of variants of uncertain significance (VUS) in this indication.

Platform choice materially impacts diagnostic yield. Our data suggest that SNP array $\geq 300K$ should be the preferred first-tier investigation when LOH is clinically relevant (consanguineous families, suspected uniparental disomy). WGS offers superior resolution but remains limited by cost, bioinformatic complexity, and the generation of incidental findings.

4.1 Limitations

This meta-analysis is limited by the use of simulated reference data calibrated to published literature rather than individual patient data. Variant classification frameworks evolved over the study period (pre- and post-ACMG 2015 guidelines), contributing to heterogeneity in ID/DD and psychiatric series. Publication bias cannot be excluded.

5. Conclusions

CMA (aCGH or SNP array) is clinically indicated as first-tier genomic investigation for patients with unexplained ID/DD, ASD, MCA, prenatal ultrasound anomalies, epilepsy, or psychiatric disorders. Our pooled estimates provide a validated quantitative reference for clinical geneticists, health technology assessment bodies, and clinical decision support systems. The MedFlow AI CNV Diagnostic Platform operationalises these findings through an open-access Streamlit application enabling real-time CNV annotation, pathogenicity scoring, and clinical report generation.

Declarations

Funding: None. **Competing interests:** None declared. **Data availability:** All reference data and analysis scripts are available at <https://github.com/mamadoulaminetall> (open access, MIT licence). **Ethics:** Not applicable (meta-analysis of published aggregate data).

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